

Ricky Faure

Principal Software Engineer

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SUMMARY

Principal Software Engineer at Optum, the technology arm of UnitedHealth Group, building enterprise-grade platform infrastructure since 2019. Deep backend and distributed systems owner: custom Kubernetes Operators in Go, Kafka-as-a-Service and Elasticsearch-as-a-Service control planes, GKE environment orchestration, and Terraform-driven infrastructure as code across on-premise and multi-cloud environments. Leads architecture design and platform management for systems that move 20+ petabytes of data and serve hundreds of enterprise teams, with hands-on SRE ownership of production streaming infrastructure spanning 800+ clusters and thousands of nodes at five nines reliability with zero data loss.

SKILLS

Core: Go, Kubernetes, Kubernetes Operators, Kubebuilder / Operator SDK, Kubernetes CRDs, Terraform, custom Terraform Provider development, GCP, GKE, Apache Kafka, Elasticsearch, Helm, CI/CD, GitHub Actions, Python, distributed systems, SRE, infrastructure as code, architecture design

Additional: Docker, Prometheus, Grafana, GitOps, bare-metal Kubernetes, GCP IAM, GCS, VPC, Artifact Registry (GAR), GCP VolumeSnapshots, certificate management, Jenkins, Java, Spring Boot, Groovy Spock, Kibana, Rally API automation, vulnerability management, mentorship and technical teaching, AI-assisted development, agentic development workflows, Claude Code, GitHub Copilot, OpenAI Codex

EXPERIENCE

Optum, UnitedHealth Group JAN 2020 — PRESENT

Principal Software Engineer, TLCP — Technology Development Program Associate (Jan 2020) → Software Engineer (Jan 2021) → Senior Software Engineer (Mar 2022) → Lead Software Engineer (Feb 2023) → Principal Software Engineer, TLCP (Feb 2024 - Present)

Empower hundreds of teams across the enterprise to build data-driven services, moving tens of petabytes of data with minimal architectural overhead, using Kubernetes Operators and CI/CD to automate deployment, management, and hosting of enterprise-grade distributed systems across on-premise and multi-cloud environments.

- Platform and team leadership: Lead 15 engineers across the Kafka-as-a-Service and Elasticsearch-as-a-Service platforms and advise 2 engineering leaders, sustaining five nines reliability through Kubernetes Operators and disciplined infrastructure-as-code practices.
- Kubernetes Operator development: Build and maintain custom Kubernetes Operators (Go, Kubebuilder) for Kafka, Elasticsearch, Prometheus, Service Monitors, GCP VolumeSnapshots, certificate management, and bare-metal Kubernetes upgrade orchestration.
- Control plane architecture: Design and manage the architecture of the platform's Kubernetes resource management control plane, in which operators generate and own concrete Kubernetes resources for automated deployment and configuration of streaming infrastructure at production scale, managing 800+ clusters across thousands of nodes in multi-tenant on-premise and GCP environments.
- Self-service marketplace migration: Achieved a 52% increase in resource deployments by migrating from GitOps to a GUI-based management system within an enterprise marketplace, backed by a custom Terraform Provider and Kubernetes resource manager.
- Cloud cost optimization: Delivered \$2.5M in annual cost savings through cloud resource optimization, instance type migrations, and elimination of excessive log retention in GCP.
- Cloud scale and GKE orchestration: Expanded the platform to handle 20+ petabytes of data movement across on-premise and GCP (GKE, GAR, IAM, GCS, VPC) by extending Kubernetes Operators for cloud environments; own

provisioning and orchestration of the 30+ GKE clusters backing these platforms through Terraform and infrastructure as code.

- SRE and production operations: Serve in on-call rotation across a fleet of 800+ clusters and thousands of nodes , delivering five nines reliability with zero customer data loss in production across the platform's history; respond to incidents, tune performance and utilization against SLAs, and stabilize high-throughput Kafka and Elasticsearch services.
- Observability as code: Converted Grafana dashboard maintenance to infrastructure as code, keeping dashboards versioned and reproducible so observability scales with the many data sources the platform has grown to.
- Agentic development workspaces: Design and build agent-driven development workspaces that standardize the team's engineering workflow into a consistent, deterministic process; encode the full breadth of the platform's distributed projects and their interdependencies into durable agent context, and author reusable agent skills that automate remediation and SRE support tasks.
- Security automation: Designed a vulnerability scanning workstream using reusable GitHub Actions, a custom Python Rally API library, and automated report generation, adopted org-wide.
- Mentorship and teaching: Mentor junior engineers in rotational programs; teach quarterly internal courses on Kubernetes Operators and Elasticsearch/Kibana.
- Featured Project: Warpstream Cluster Provisioning Platform (Q4 2025 - Q1 2026)
- Architected the end-to-end design for Warpstream-based Kafka cluster provisioning: the networking design, the operator, agent, and resource specifications, and the resource flows from the customer-facing resource manager down to the backend infrastructure.
- Extended the platform's Kubernetes operators for net-new resource kinds and delivered a net-new Warpstream operator (Go) that generates all required Kubernetes resources and connects them into Cloud Storage buckets, Warpstream API registrations, and agent configs.
- Co-designed the Terraform cloud infrastructure for the Warpstream Clusters-as-a-Service deployment, including GCP Cloud Storage provisioning, VPC configuration, DNS, and IAM.
- Shipped to two of Optum's largest Apache Kafka on GCP customers as beta; Warpstream's diskless, Cloud Storage-based architecture eliminates local disk I/O from the stream processing data path, projected to reduce their annual Kafka infrastructure spend by approximately 80% .
- Featured Project: Kafka Cluster Provisioning Platform on GCP
- Build and operate the Kafka-as-a-Service provisioning platform on GCP: custom Kubernetes Operators (Go) own the full cluster lifecycle on GKE, integrated with GCS, IAM, VPC, and Artifact Registry through Terraform-managed cloud infrastructure.
- Delivered provisioning first through GitOps-driven CI/CD pipelines, then migrated it to GUI-based self-service in the enterprise marketplace developer platform, backed by the custom Terraform Provider and Kubernetes resource management control plane.
- Featured Project: Kafka Cluster Provisioning Platform on Bare-Metal
- Build and operate the on-premise Kafka-as-a-Service platform on bare-metal Kubernetes: operators automate cluster provisioning, configuration, certificate management, and rolling upgrades across multi-tenant environments, delivered first through CI/CD pipelines and subsequently through the self-service developer platform.
- Keep streaming workloads available through node cycling via the custom bare-metal Kubernetes upgrade orchestration operator.
- Featured Project: Elasticsearch Provisioning Platform on GCP
- Extended the Elasticsearch-as-a-Service operators into GCP, provisioning clusters on GKE with Terraform-managed cloud infrastructure and operator-driven GCP VolumeSnapshot backup and restore; provisioning ran first through CI/CD pipelines before migrating to the self-service developer platform.
- Featured Project: Elasticsearch Provisioning Platform on Bare-Metal
- Build and operate Elasticsearch-as-a-Service on bare-metal Kubernetes: operators automate cluster provisioning, configuration, certificate management, and Kibana access, with Prometheus and Service Monitor operators providing observability across the fleet; provisioning moved from CI/CD pipelines to the self-service developer platform.

Technologies: Go, Kubebuilder, Kubernetes Operators, Helm, Terraform, GCP, GKE, GitHub Actions, Python, Kafka, Warpstream, Elasticsearch, Prometheus, Jenkins, Docker

TDP Software Development Intern

- Returned for three consecutive summer internships (June - August, 2017, 2018, and 2019) with similar roles and responsibilities each year.
- Contributed business-critical features to a monolithic application, complete with feature implementations and automated unit and integration testing.

Technologies: Java, Spring Boot, Groovy Spock

PROJECTS

**Yo-Yo Mount Visualizer
(active development)**

TypeScript · React · react-three-fiber · 3D visualization · graph modeling ·
pathfinding · physics simulation · Zod schema validation · Vite · pnpm
monorepo · vitest

3D trick engine that models yo-yo string mounts as graph topologies and aims to discover new tricks through pathfinding. Encodes mounts as ordered, schema-validated string traversals with canonical serialization and hash-based deduplication, renders them in an interactive 3D visualizer with switchable cameras and timeline scrubbing, and animates transitions with a fixed-timestep Verlet rope physics simulation.

Spin Ledger (closed source)

Python · FastAPI · SQLAlchemy 2.0 · Alembic · Typer CLI · REST API
design · database migrations · rule-based entity extraction · data
normalization · API-driven scraping · self-hosted deployment

Single pane of glass for buy/sell/trade activity across skill toy community forums. Ingests unstructured, schemaless forum listings that resist conventional parsing, normalizes them into a taxonomy-aware data model, and serves them through an indexed, searchable marketplace dashboard, turning fragmented community commerce into structured, queryable data. Ingestion runs against Discourse's JSON API with deterministic rule-based entity extraction that exploits the forums' strict listing conventions, enriched by live lookups against a companion taxonomy catalog service. LLM-based parsing grounded in the live taxonomy is in development, deliberately deferred at launch where it was cost-prohibitive for a project at this stage.

Self-Hosted Home Lab

Docker · Caddy · reverse proxy · TLS · DNS · VPN tunneling · NAS · Linux ·
networking · self-hosting · infrastructure administration

Multi-machine home lab orchestrating a fleet of containerized services with Docker, fronted by Caddy as a reverse proxy with automatic HTTPS; every service is served exclusively over TLS, inside and outside the network. Hosts a media server, network-attached storage, network-wide DNS-sinkhole ad and tracker blocking, credential and secrets vaults, and deployments of closed-source personal projects including Spin Ledger. Private encrypted tunnels provide secure remote access to the network without exposing services publicly.

OSS Feed

JavaScript · GitHub Actions · RSS/Atom · GitHub Pages · YAML

Self-hosted weekly RSS digest that tracks releases from curated open source projects. A watchlist YAML feeds GitHub Actions to fetch releases, generate an XML feed, and publish via GitHub Pages, with Slack/Discord webhook notifications and state management to avoid duplicate alerts.

Rubik's Cube Genetic Algorithm Solver

Python · genetic algorithms · matrix math

Evolutionary algorithm that solves a fully simulated Rubik's Cube model, exploring genetic computation applied to combinatorial optimization.

SafetyNet

Swift · iOS · P2P networking · E2E encryption

iOS peer-to-peer messaging app requiring no Wi-Fi or cellular data. Uses device-to-device networking with end-to-end encryption for resilient off-grid communication.

Ray Tracing Engine

JavaScript · graphics pipeline · 3D rendering · Blinn-Phong

Browser-based ray tracer in vanilla JavaScript. Renders spheres and triangles with Blinn-Phong lighting, reflections, and a free-moving camera, no GPU required.

Sheet Vision (Senior Design)

ElectronJS · ReactJS · AWS · Python · computer vision

Application that reads sheet music, plays it back, and listens to the user in real time, providing feedback to help learners draw parallels between notation and sound.

Run Samurai, Run!

P5.js · HTML/CSS · PHP · AJAX · game loop development

Top-down 2D infinite side-scrolling runner for mobile. Players fight through enemy mobs for upgrades and high scores, with single-player and head-to-head multiplayer modes.